# Global Ratio Monotonicity for a Killed von Mises Bridge

[COMPLETE-THEOREM MANUSCRIPT — SEAL IN PROGRESS; DO NOT SUBMIT.
Pending relay lemmas are marked [SLOT].]

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July 2026

**Abstract**

For $\beta > 0$ let $I_m = I_m(\beta)$ denote modified Bessel functions of the first kind, and set $a_m = I_m^2\,[(m-1)I_{m-1}^2 + (m+1)I_{m+1}^2]$, $b_m = m\,I_m^4$, $F_A(t) = \sum_{m\geq 1} a_m \sin(mt)$, $F_B(t) = \sum_{m\geq 1} b_m \sin(mt)$, and $E = F_A/(2F_B)$. The global ratio-monotonicity problem for the surface expansion of a two-dimensional $SU(2)$ lattice gauge observable is: (i) $F_B > 0$ on $(0, \pi)$, and (ii) $E' < 0$ on $(0, \pi)$, for every $\beta > 0$. We prove (i) twice, and prove (ii) for $0 < \beta \leq 3$ via an exact pair-identity skeleton plus an archived interval certificate executed by two independent interval-arithmetic implementations. The structural core is exact: $E$ is, as an algebraic identity, the mean of $\cos\psi$ under the midpoint law of a four-step killed von Mises bridge; the generating kernels reduce, via the Neumann addition theorem, to two-dimensional integrals of a *single* Bessel function whose saddle deficit is an exact sum of two squares; and exact saddle cancellations yield the coefficients of the (verified) closed second-order law $E = \cos(t/2)(1 - c(t)/\beta) + O(\beta^{-2})$, $c(t) = (4\cos^2(t/4) - 1)/(2\cos(t/4)\cos(t/2))$. Three certified negative results (interval arithmetic, two implementations, nested enclosures) kill every monotone full-path coupling, with an exact mechanism at threshold $\beta|\cos t| = 3/2$. For the remaining range we give a two-scale closure map with a verified candidate threshold $\beta_0 = 15$ and, at the $\pi$-endpoint, exact identities for the cubic coefficient $c_3$ (telescoped alternating form, integral form, parity) together with its verified prefactor law. Every claim is labelled exact / certified / verified; the machine-checked lemmas are Lean 4/Mathlib, machine-checked modulo classical Bessel inputs carried as named hypotheses.

## 1 Introduction

This paper consolidates the resolution status of a two-part positivity problem that arises as the last analytic gap in a surface (character) expansion for two-dimensional $SU(2)$ lattice gauge theory [5, 6, 7]. The problem is stated entirely in terms of modified Bessel functions and trigonometric series, and we treat it as such; no claim of physical consequence beyond the stated expansion is made, and importance is not inherited by thematic proximity.

Throughout, $\beta > 0$, $I_m = I_m(\beta)$, and

$$a_m = I_m^2\big[(m-1)I_{m-1}^2 + (m+1)I_{m+1}^2\big], \qquad b_m = m\,I_m^4, \qquad m \geq 1. \tag{1}$$

Both sequences decay super-exponentially, so

$$F_A(t) = \sum_{m\geq 1} a_m \sin(mt), \qquad F_B(t) = \sum_{m\geq 1} b_m \sin(mt), \qquad E(t) = \frac{F_A(t)}{2F_B(t)}, \tag{2}$$

are real-analytic on $\mathbb{R}$, and the Wronskian $W = 2(F_A' F_B - F_A F_B')$ satisfies $W = 4F_B^2\,(E)'$ wherever $F_B \neq 0$ [7].

**Conjecture 1** (global ratio monotonicity)**.** *For every $\beta > 0$:* (i) $F_B > 0$ *on* $(0, \pi)$*;* (ii) $E' < 0$ *on* $(0, \pi)$*.*

Part (i) is proved below (twice). Part (ii) is proved for $0 < \beta \leq 3$ (Theorem 26; exact skeleton plus the archived certificate `certify_thmB.py` and its Arb twin), is supported by an exact structural reduction and a verified truth map for $\beta \geq 15$, and is otherwise open; Section 9 states the precise two-scale closure map. The conjecture was born inside this programme (it is the global form of an $\varepsilon$-window positivity required by the surface expansion); no external pedigree is claimed.

*Convention* 2 (tricotomy)*.* Every assertion below carries one of three labels. *Exact*: a proved identity or inequality. *Certified*: a numerical statement rigorously enclosed by interval arithmetic with outward rounding, together with complete analytic bounds for domains and truncations. Precision-stability re-runs and independent implementations are *software audits*, recorded separately with their declared coverage: the certified negatives `certify_time3_negative` and `certify_bridge_matrix` carry a *full twin audit* (nested re-computation plus a complete independent Arb implementation, both transcripts archived); for `certify_thmB` both implementations exist and the Arb transcript is archived — its row is *amber* only because the `mpmath.iv` canonical transcript is still in progress; the $[6, 15]$ coverage will carry a *planned pre-registered sampled independent audit* (Remark 20). *Verified*: high-precision numerics with stated grids and precision policies, not a proof. No verified statement is used as a hypothesis of an exact one.

## 2 The theorem and the relay

**Theorem 3** (the Surface Theorem; [**SLOT: seal completion**])**.** *For every $\beta > 0$:* (i) $F_B > 0$ *on* $(0, \pi)$*, and* (ii) $E' < 0$ *on* $(0, \pi)$*.*

Part (i) is Theorem 23. Part (ii) is proved by the following relay. Rows marked [**SLOT**] are pending; rows marked *amber* have one archived witness with the second in progress; all other rows are complete with the stated witnesses.

| range | method | artifact | transcript | state |
|---|---|---|---|---|
| $(0, \frac{1}{20}]$ | small-$\beta$ lemma | Lem. 27 | — | complete |
| $[\frac{1}{20}, 3]$, all $t$ | pair skeleton | `certify_thmB` | Arb | *amber* |
| $[3, 6]$, $t \in [0.6, \pi - \frac{1.5}{\beta}]$ | series intervals | `certify_bulk` | Arb | *amber* |
| $[6, 15]$, $t \in [0.6, \pi - \frac{1.5}{\beta}]$ | exp-factored integrator | `exp_integrator` | [**SLOT**] | local box: Prop. 19 |
| $[15, \infty)$, bulk | analytic relay | [**SLOT**] | — | pending |
| $t \in (0, 0.6]$, $\beta \geq 3$ | sinc + minor criterion | [**SLOT: sinc-cert**] | — | design ratified |
| $t \in [\pi - \frac{C_{\mathrm{win}}}{\beta}, \pi)$ | $c_3$ + radius | [**SLOT: 4 const.**] | — | $A$ verified |

Splice boundaries are explicit and overlapping: $\beta = 3,\ 6,\ 15$; $t = 0.6$ and $t = \pi - C_{\mathrm{win}}/\beta$, with $C_{\mathrm{win}} = 1.5$ a candidate pending $C_{\mathrm{mirror}} \leq 1.5 \leq C_{\mathrm{quad}}$ (the moving-boundary constant is always written $C_{\mathrm{win}}$; the bare letter $C$ means $\cos(t/2)$). The remainder of this manuscript is the completed material, verbatim from the consolidation paper.

*Remark* 4 (supersession of the $e_4$ route). An earlier plan closed the block $t \in (0, 0.6]$, $\beta \geq 3$ by an $e_4$-coefficient lemma with a fixed radius; it is superseded, not silently deleted. A feasibility check showed the sinc criterion below dominating with ratios from 15 up to $10^{80}$ across $\beta \in [3, 15]$ for $t \leq 0.3$ (*verified*, design-grade), while the apparent infeasibility of $(0, 0.6]$ belonged to the cancellation of the series tool, which the exp-factored integrator does not suffer. The ratified route has three components: the exact sinc positivity (Lemma 5), cheap one-dimensional-in-$\beta$ minor-criterion certificates on $(0, t_*(\beta)]$, and the extension of the integrator strip from $t = 0.6$ down to $t_*(\beta) \approx 0.45$ with splice overlap. The slot's pending content is the full minor criterion with its uniform overlap; the $e_4$ lemma is retired.

**Lemma 5** (exact; sinc positivity)**.** *Let $1 \leq m < n$ and $T_{mn}(t) = (m - n)\sin((m + n)t) + (m + n)\sin((n - m)t)$ as in Section 4. Then $T_{mn}(t) > 0$ for $0 < t \leq \pi/(m + n)$.*

*Proof.* Write $s = m + n$, $p = n - m \geq 1$, so $T_{mn}(t) = s\sin(pt) - p\sin(st)$. Since $\sin(x)/x$ is strictly decreasing on $(0, \pi]$ and $0 < pt < st \leq \pi$, we get $\sin(pt)/(pt) > \sin(st)/(st)$, i.e. $st\sin(pt) > pt\sin(st)$; dividing by $t > 0$ gives $s\sin(pt) > p\sin(st)$. □

**Lemma 6** (relay criterion for $\beta \in [6, 15]$)**.** *Let $\langle f\rangle = \iint_{[-\pi,\pi]^2} Kf\,ds\,d\alpha$ with $K = I_1(2\beta R)/R > 0$, and let $\bar{E}$ be any rational constant. Then $E' < 0$ at $(t, \beta)$ if and only if $\langle D\rangle > 0$ and*

$$W_c := \langle N_t\rangle\langle D\rangle + \langle (N - \bar{E}D)\,\partial_t\log K\rangle\langle D\rangle - (\langle N\rangle - \bar{E}\langle D\rangle)\langle D\,\partial_t\log K\rangle < 0$$

*(the $\bar{E}$-terms cancel identically). The certificate encloses the five integrals by exp-factored interval cells: $z = 2\beta R$ is linearized at the exact rational cell center with interval-gradient remainder; the linear exponential integrates exactly; and the two scaled Bessel factors are $A(z) = e^{-z}I_1(z)/z$ — strictly decreasing, since $(\log A)' = I_0/I_1 - 1 - 2/z < 0$ follows from the classical lower bound at $\nu = 0$ (Amos [1]; [2], Thm. 1) and $\sqrt{1 + z^2} \leq z + 1$ — enclosed by pure endpoint evaluation, and $B(z) = e^{-z}H_B(z)$, where*

$$H_B(z) := \frac{I_0(z) - 2I_1(z)/z}{z^2}, \qquad H_B(0) = \tfrac{1}{8}$$

*(positive-coefficient entire series; the subscript distinguishes it from the entry-law kernel $H(\psi)$), enclosed unconditionally by $[e^{-z_{\mathrm{hi}}}H_B(z_{\mathrm{lo}}), e^{-z_{\mathrm{lo}}}H_B(z_{\mathrm{hi}})]$; the identity $K\,\partial_t\log K = 32\beta^3 c\,c'\,(1 - P - Q)H_B(z)$, $c = \cos(t/4)$, removes $I_1(z)/z$ from the covariance integrands exactly. The $(t, \beta)$-box layer is Proposition 19; the global $[6, 15]$ coverage remains the pending content of the relay row.*

**Lemma 7** (exact; the $S$-free relative criterion)**.** *Let $S = \sin(t/2)$, $E = \langle N\rangle/\langle D\rangle$, and suppose $\langle D\rangle > 0$. Then for all $t \in (0, \pi)$ and $\beta > 0$,*

$$\frac{E'(t, \beta)}{-\frac{1}{2}S} = \frac{\langle \cos 2s + \cos\alpha\cos s\rangle + 8\beta^3\iint (N - E\,D)(1 - P - Q)\,H_B(z)\,ds\,d\alpha}{\langle D\rangle},$$

*and consequently*

$$E' < -\tfrac{1}{4}S \iff \mathcal{C} := \langle \cos 2s + \cos\alpha\cos s\rangle + 8\beta^3\iint (N - E\,D)(1 - P - Q)H_B - \tfrac{1}{2}\langle D\rangle > 0.$$

*Proof.* Since $D_t = 0$, $E' = [\langle N_t\rangle + \iint (N - ED)\,\partial_t K]/\langle D\rangle$. The first term factors $S$ exactly: $N_t = -\frac{S}{2}(\cos 2s + \cos\alpha\cos s)$. So does the second: $\partial_t K = K\,\partial_t \log K = 32\beta^3 c\,c'\,(1-P-Q)H_B(z)$ with $c = \cos(t/4)$, and $c\,c' = \cos(t/4)\cdot\big(-\frac{1}{4}\sin(t/4)\big) = -\frac{S}{8}$ by the double angle, so $\partial_t K = -4\beta^3 S\,(1-P-Q)H_B$. Dividing by $-\frac{1}{2}S$ cancels $S$ identically and yields the display. For the equivalence: the denominator $-\frac{1}{2}S$ is negative on $(0,\pi)$, so $E' < -\frac{1}{4}S$ is the *lower* bound $E'/(-\frac{1}{2}S) > \frac{1}{2}$, i.e. $\mathcal{C} > 0$. □

**Corollary 8** (exact; the bilinear quotient-free form)**.** *With* $\Phi := \cos 2s + \cos\alpha\cos s$ *(the letter* $A$ *stays reserved for the scaled Bessel factor* $e^{-z}I_1(z)/z$*),* $J_N := \iint N(1-P-Q)H_B\,ds\,d\alpha$*,* $J_D := \iint D(1-P-Q)H_B\,ds\,d\alpha$*, and* $\langle D\rangle > 0$*,*

$$E' < -\tfrac{1}{4}\sin(t/2) \iff \widehat{\mathcal{C}} := \langle D\rangle\langle\Phi\rangle + 8\beta^3\big(\langle D\rangle J_N - \langle N\rangle J_D\big) - \tfrac{1}{2}\langle D\rangle^2 > 0.$$

*Proof.* Multiply $\mathcal{C}$ of Lemma 7 by $\langle D\rangle > 0$ and substitute $E = \langle N\rangle/\langle D\rangle$: the quotient disappears by bilinearity, and the inequality direction is preserved. $\widehat{\mathcal{C}}$ is a bilinear combination of five integrals with no quotient anywhere; it is the assembly's sole target. □

*Remark* 9 (why $\widehat{\mathcal{C}}$ is the assembly's object). In $\widehat{\mathcal{C}}$ the variable $t$ enters only through $c = \cos(t/4) \in (\frac{\sqrt{2}}{2}, 1)$: no kinematically forced vanishing factor and no singularity anywhere in the relay range — the common linear zero of margin and correction at $t \to 0$ has been cancelled identically. The endpoint degeneracy is thus removed structurally; uniform positivity still requires explicit uniform estimates, which are the assembly's content. The first-order calibration constant is $a(t) = [\frac{1}{2} + \frac{1}{8c^2}]/c \le \frac{3\sqrt{2}}{4} \approx 1.061$ (supremum at $c = \sqrt{2}/2$), so the first-order threshold is $2\sup a \approx 2.12$; the target of the analytic relay is $\widehat{\mathcal{C}} > 0$ uniformly for $\beta \ge \beta_0$ with the remainder included. At the calibration point $(t,\beta) = (1.5, 8)$ the measured value is $\mathcal{C}/\langle D\rangle \approx 0.417$ (*verified*), against the threshold 0. The built-in unit test: the $1/\beta$ expansion of $\mathcal{C}/\langle D\rangle + \frac{1}{2}$ must reproduce $1 - a(t)/\beta$; the audited quantity is $\beta\cdot$(deviation from first order), which must converge (an $O(\beta^{-2})$ remainder halves the $\beta$-*scaled* deviation when $\beta$ doubles, and divides the unscaled deviation by four).

**Lemma 10** (exact; far-region pointwise and kernel bounds)**.** (a) $\min_{s,\alpha}\Phi = -\frac{9}{8}$ *exactly: with* $u = |\cos s|$ *the minimum of* $2u^2 - u - 1$ *on* $[0,1]$ *is attained at* $u = \frac{1}{4}$ *and equals* $-\frac{9}{8}$*.* (b) $|D| \le 2$ *and, by Corollary 30* ($|E| < 1$)*,* $|N - ED| \le |N| + |D| \le (2|C|+1) + 2 \le 5$ *and* $|1-P-Q| \le 1$*.* (c) *On the deficit region* $\{(s,\alpha) : 2c - R \ge \Delta_0\}$ *one has* $z = 2\beta R \le z_s - 2\beta\Delta_0$ *with* $z_s = 4\beta c$*, and since* $A(z) = e^{-z}I_1(z)/z$ *is strictly decreasing with* $A(0^+) = \frac{1}{2}$*,*

$$K = 2\beta\, e^z A(z) \;\le\; \beta\, e^{z_s}\, e^{-2\beta\Delta_0} \quad \text{on the region.}$$

*The far-region contributions of all five integrals of Corollary 8 are then bounded by the two tail families of Lemma 11 below.*

*Proof.* (a) $\cos\alpha\cos s \ge -|\cos s|$ with equality attainable, and $\cos 2s = 2u^2 - 1$; minimizing $g(u) = 2u^2 - u - 1$ gives $g(\frac{1}{4}) = -\frac{9}{8}$. (b) Triangle inequalities; $|N| \le |C|\,|\cos 2s| + |\cos\alpha|\,(|C\cos s| + \sin^2 s) \le 2|C| + 1$. (c) $R \le 2c - \Delta_0$ gives $z \le z_s - 2\beta\Delta_0$; write $K = I_1(z)/R = 2\beta I_1(z)/z = 2\beta e^z A(z)$ and use the monotonicity of $A$ (Section on the two scaled functions) with $A(z) \le A(0^+) = \frac{1}{2}$ uniformly. □

**Lemma 11** (exact; the two tail families)**.** $0 < H_B(z) \le \frac{1}{4}I_1(z)/z = \frac{1}{4}e^z A(z) \le \frac{1}{8}e^z$ *(term by term: the coefficient of* $(z/2)^{2k}$ *in* $H_B$ *is* $\frac{1}{4}\frac{1}{(k+2)\,k!\,(k+1)!} \le \frac{1}{8\,k!\,(k+1)!}$*, the coefficient in* $\frac{1}{4}I_1(z)/z$

*being* $\frac{1}{8\,k!\,(k+1)!}$*). Consequently, writing* $\delta X$ *for the far-region part of an integral* $X$ *over* $\{2c-R \geq \Delta_0\}$*,*

$$|\delta\langle f\rangle| \leq 4\pi^2\,|f|_\infty\,\beta\,e^{z_s-2\beta\Delta_0}, \qquad |\delta J_f| \leq \tfrac{\pi^2}{2}\,|f(1-P-Q)|_\infty\,e^{z_s-2\beta\Delta_0},$$

*for the three* $K$*-integrals and the two* $H_B$*-integrals respectively; on the same region* $z \leq z_s - 2\beta\Delta_0$ *also bounds* $e^z$ *in* $H_B \leq \frac{1}{8}e^z$*.*

**Proposition 12** (exact; Region III propagation)**.** *Let* $\widehat{\mathcal{C}}_{\mathrm{near}}$ *denote* $\widehat{\mathcal{C}}$ *with every integral restricted to* $\{2c - R < \Delta_0\}$*. Then for every* $\Delta_0 \in (0, 2c)$ *and* $\beta \geq 1$*,*

$$|\widehat{\mathcal{C}} - \widehat{\mathcal{C}}_{\mathrm{near}}| \;\leq\; 576\,\pi^4\beta^4\,e^{2z_s}\,e^{-2\beta\Delta_0}.$$

*Proof.* Global masses: $\iint K \leq 4\pi^2\beta e^{z_s} =: M$ and $\iint H_B \leq \frac{\pi^2}{2}e^{z_s} =: M_H$ (the pointwise bounds $K \leq \beta e^{z_s}$, $H_B \leq \frac{1}{8}e^{z_s}$ hold on all of $[-\pi,\pi]^2$ since $z \leq z_s$). With the pointwise constants of Lemma 10 ($|\Phi|_\infty \leq 2$, $|D|_\infty \leq 2$, $|N|_\infty \leq 3$, $|N(1-P-Q)|_\infty \leq 3$, $|D(1-P-Q)|_\infty \leq 2$), Lemma 11 gives $|\delta\langle D\rangle|, |\delta\langle\Phi\rangle| \leq 2a$, $|\delta\langle N\rangle| \leq 3a$, $|\delta J_N| \leq 3b$, $|\delta J_D| \leq 2b$, where $a := Me^{-2\beta\Delta_0}$, $b := M_He^{-2\beta\Delta_0}$ (note $4\pi^2\beta e^{z_s-2\beta\Delta_0} = a$ and $\frac{\pi^2}{2}e^{z_s-2\beta\Delta_0} = b$). For any product, $|XY - X_{\mathrm{n}}Y_{\mathrm{n}}| \leq |\delta X|\,|Y|_{\max} + |X|_{\max}|\delta Y|$ with $|\langle f\rangle| \leq |f|_\infty M$, $|J_f| \leq |f(1-P-Q)|_\infty M_H$. Summing: $\langle D\rangle\langle\Phi\rangle$ contributes $\leq 8aM$; $\langle D\rangle J_N$ and $\langle N\rangle J_D$ contribute $\leq 12MM_He^{-2\beta\Delta_0}$ each (using $aM_H = bM = MM_He^{-2\beta\Delta_0}$); $\frac{1}{2}\langle D\rangle^2$ contributes $\leq 4aM$. Total: $12aM + 192\beta^3MM_He^{-2\beta\Delta_0} = (192\pi^4\beta^2 + 384\pi^4\beta^4)e^{2z_s}e^{-2\beta\Delta_0} \leq 576\pi^4\beta^4e^{2z_s}e^{-2\beta\Delta_0}$ for $\beta \geq 1$. □

*Remark* 13 (the choice of $\Delta_0$ belongs to the assembly)*.* With these deliberately crude (but fully named) constants, at $\beta = 15$ the bound forces $\Delta_0 \approx 1.2$ rather than the design value 0.3: the natural saddle-mass scale is $\langle D\rangle \asymp e^{z_s}\beta^{-3/2}$ (saddle height $\asymp e^{z_s}\beta^{-1/2}$ times a two-dimensional Gaussian area $\asymp \beta^{-1}$), so $\langle D\rangle^2 \asymp e^{2z_s}\beta^{-3}$ and the comparison quotient carries $\beta^7e^{-2\beta\Delta_0}$, consuming $\approx 34$ of exponent at $\beta = 15$. By the uniform saddle-mass Lemma 16 ($m_* \geq \frac{1}{2}$), $\Delta_0 = 1.2$ suffices with margin (Corollary 14). The enlargement changes no leading bound, subject to the forthcoming geometric shell estimate: the shell technique that will bound Region II is insensitive to its outer edge — the telescoping sum over deficit shells is controlled by the kernel mass beyond the *inner* radius $\Delta_1$ (up to its geometric closing factor, part of that estimate). Since $\Delta_0 < 2c$ is free, the constant hunt stays honest.

**Corollary 14** (Region III numeric tooth)**.** *By Lemma 16* ($m_* \geq \frac{1}{2}$)*, for* $\beta \geq 15$ *and* $t \in (0, \pi - C_{\mathrm{win}}/\beta]$*, with* $\Delta_0 = 1.2$*,*

$$\frac{|\widehat{\mathcal{C}} - \widehat{\mathcal{C}}_{\mathrm{near}}|}{\langle D\rangle^2} \;\leq\; \frac{576\pi^4}{m_*^2}\,\beta^7e^{-2.4\beta} \;\leq\; 2304\,\pi^4\,\beta^7e^{-2.4\beta},$$

*and the right side is decreasing in* $\beta$ *on* $[15,\infty)$ ($\frac{d}{d\beta}\log(\beta^7e^{-2.4\beta}) = \frac{7}{\beta} - 2.4 < 0$ *for* $\beta \geq 15$)*, with value* $\leq 8.9 \times 10^{-3}$ *at* $\beta = 15$*: a single evaluation certifies the whole half-line.* (*An earlier draft hypothesized* $m_0 = 1$*, consistent with every measurement but not reached by the proven floor; it is superseded by Lemma 16, not falsified — constant four times uglier, foundation harder.*)

*Convention* 15 (cited constants travel with their normalization)*.* All double integrals in this section are over $[-\pi,\pi]^2$ with no normalization: $\langle f\rangle := \iint Kf\,ds\,d\alpha$. Every numerically measured constant quoted in this manuscript travels with its triplet (normalization, $t$, $\beta$); the saddle-mass lemma is stated as an explicit function $m_0(c)$, never as a loose scalar, since the saddle prefactor carries a $c$-dependence and measurements confirm it ($m_0 \approx 1.492$ at $(t,\beta) = (1.5, 15)$, 2.110

at $(2.5, 15)$, 1.353 at $(1.0, 40)$ under this convention). The one historical discrepancy is *closed exactly*: via $F_B = \frac{\beta \sin(t/2)}{16\pi^2}\langle D\rangle$, i.e. $\langle D\rangle = 16\pi^2 F_B/(\beta\sin(t/2))$, the $F_B$-series reproduces all three readings; an earlier report of 1.05 was the $(2.5, 15)$ value computed with the bracket $D/2$ in place of $D$ — a factor of exactly 2, identified, not a numerical uncertainty. Uniform floors are written $m_* := \inf_{c\in[\sqrt{2}/2,1]} m_0(c)$.

**Lemma 16** (uniform saddle mass). *For $\beta \geq 15$ and $c = \cos(t/4) \in [\frac{\sqrt{2}}{2}, 1]$,*

$$m_0(c) \;=\; \langle D\rangle\, e^{-z_s}\beta^{3/2} \;\geq\; \frac{9\pi}{10\,c^2\sqrt{8\pi c}}\Big(1 - \frac{3}{20\beta c}\Big)\Big(1 - e^{-\beta c/5}\Big)^2 \;\geq\; \frac{1}{2},$$

*so $m_* \geq \frac{1}{2}$ on the whole relay range.*

*Proof outline, five one-line links.* Restrict to the rectangle $|s|, |\alpha| \leq 1/\sqrt{5}$ (Gaussian window $\sigma^2 = \frac{1}{5}$). There: (i) $D = \cos s + \cos\alpha \geq 2\cos(1/\sqrt{5}) \geq \frac{9}{5}$; (ii) the deficit obeys the quadratic lower bound $4c^2 - R^2 \geq (P+Q)(4c^2 - (P+Q))$, oriented by Lemma 17(a); (iii) the one-term lower bound of the $L2'$ family (re-verified on $z \in [10, 120]$) bounds $I_1(z) \geq e^z(1 - \frac{3}{8z} - \cdots)/\sqrt{2\pi z}$ from below at $z \geq z_s(1 - \frac{1}{10})$; (iv) the truncated Gaussian integrals are exact, contributing the factor $\frac{\pi}{\beta c}(1 - e^{-\beta c/5})^2$ and the emergent $\beta^{-3/2}$; (v) collecting constants gives the display, whose infimum over $c \in [\frac{\sqrt{2}}{2}, 1]$ at $\beta = 15$ is attained at $c = 1$ with value $0.504 \geq \frac{1}{2}$ (at $c = \frac{\sqrt{2}}{2}$ the value is 1.025). Design-grade check: the true mass exceeds this bound by a stable factor 2.2–2.8 across six cells — the healthy signature of a crude-but-honest floor. □

**Lemma 17** (exact; two global mini-lemmas for Step 0). (a) *If $c^2 \geq \frac{1}{2}$ (the whole relay range) then $R \leq 2c$, i.e. $z \leq z_s$, on all of $[-\pi,\pi]^2$: indeed $4c^2 - R^2 = 4c^2(P+Q) - 4PQ \geq 8c^2\sqrt{PQ} - 4PQ = 4\sqrt{PQ}\,(2c^2 - \sqrt{PQ}) \geq 0$ since $\sqrt{PQ} \leq 1 \leq 2c^2$.* (b) *The fluctuation $F := N - CD$ satisfies the global pointwise bound $|F| \leq 12P$: from $\cos 2s - \cos s = -2\sin(3s/2)\sin(s/2)$ and $1 - \cos s = 2\sin^2(s/2)$,*

$$F = -2C\sin(3s/2)\sin(s/2) - \cos\alpha\big[2C\sin^2(s/2) + \sin^2 s\big],$$

*and $|\sin(3s/2)| \leq 3|\sin(s/2)|$, $\sin^2 s = 4P(1-P) \leq 4P$, $|C| \leq 1$ give $|F| \leq 6P + (2P + 4P) = 12P$.*

*Remark* 18 (Step 0 of the assembly: the $E$-window). Lemma 17 makes the first assembly step a single moment estimate: $|E - C| = |\langle F\rangle|/\langle D\rangle \leq 12\langle P\rangle_K/\langle D\rangle$, with no case split (the $F$-bound is global, and (a) orients every kernel bound in one direction on the whole torus). The pending content is the moment bound $\langle P\rangle_K/\langle D\rangle \leq [4\beta c]^{-1}(1 + \varepsilon(\beta, c))$ with explicit $\varepsilon$, yielding $|E - C| \leq \frac{3}{\beta c}(1+\varepsilon)$ — crude constant $3/c$ against the exact first-order $\kappa(c) = (4c^2-1)/(2c) \leq \frac{3}{2}$ (factor 2–4 of slack, sufficient for its sole client: $\mathrm{num}_2$ needs $|C - E| = O(1)/\beta$ with a named constant). The eight-cell calibration table ($\beta|E - C| \nearrow \kappa(c)$ from below) is the step's unit test.

**Proposition 19** (certified; local box: $(t, \beta) \in [1.5, 1.51]\times[8, 8.05]$). *For every $(t, \beta) \in [1.5, 1.51]\times[8, 8.05]$ one has $\langle D\rangle > 0$ and $W_c < 0$; hence, by Lemma 6, $E'(t, \beta) < 0$ on this box.*

*Certificate.* Ball-arithmetic machine certificate (python-flint 0.9.0 / Arb). The executed script is identified by SHA-256 prefix `1d888e99` and is archived byte-exact in the repository, together with the self-contained transcript (both file names carry the hash; the transcript's provenance header records the full hash, library versions, and stage parameters). Three stages, one transcript. *Point*

$(1.5, 8)$: 1,951,141 cells, $W_c/\langle D\rangle^2 \subset [-0.4546, -0.1752]$. *Stability* (precision 120, $\Delta z \leq 0.12$): 2,500,336 cells, ratio $\subset [-0.4304, -0.1978]$, nested inside the point enclosure. *Box*: union of nine parametric sub-boxes ($\Delta\beta \leq 0.02$, $\Delta t \leq 0.005$), each with strictly negative $W_c$ enclosure and $\langle D\rangle > 0$ (worst $W_c$ upper end $\leq -2.8 \times 10^{21}$), 22,986,387 cells in total, 7558 s. All comparisons are strict ball comparisons; the displayed intervals are outward-rounded supersets of the printed enclosures. This proposition validates the mechanism and calibrates the coverage design; it does *not* close the $[6, 15]$ row, which stays open until the full region $6 \leq \beta \leq 15$, $0.6 \leq t \leq \pi - 1.5/\beta$ is covered. □

*Remark* 20 (what certifies, and what audits). The mathematical certificate for each sub-box is its *single* rigorous Arb enclosure: outward-rounded ball arithmetic makes one honest enclosure a proof of the strict inequalities, per sub-box. The precision-120 stability stage (with its nesting inside the point enclosure) and the *planned* pre-registered sampled `mpmath.iv` reproduction are *implementation audits*: they test the software that produced the enclosures, not the inequalities themselves, and they are declared as such in the relay table. This fixes the convention for the $[6, 15]$ coverage: one certificate-grade Arb enclosure per sub-box; stability and sampling audit the pipeline.

**Lemma 21** (safe square roots in the archived run)**.** *In the archived script (*SHA-256 `1d888e99`*), the square-root guard clamps a negative lower endpoint to* 0 *and, when the* upper *endpoint ball straddles* 0*, returns the dyadic upper bound $2^{8-p/2}$ at working precision $p \geq 90$. This bound is valid throughout the run. Indeed, the guarded inputs are true squares: $R^2 = 4[c_0^2(1-P-Q)+PQ]$ with $c_0^2 \leq 1$, $P, Q \in [0, 1]$, and $z^2 = 4\beta^2 R^2$ with $\beta \leq 15$, so $|R^2| \leq 16$, $|z^2| \leq 14400$; the corresponding* balls *are built by at most twelve correctly rounded operations on factors of unit size, hence $|R^2_{\text{ball}}| < 16.01$ and $|z^2_{\text{ball}}| < 14401 < 2^{14}$ (magnitude means $|\text{mid}| + \text{rad}$). The upper-endpoint ball $u = \text{mid} + \text{rad}$ involves two exact arf-to-ball conversions and one correctly rounded addition, so $\text{rad}(u) \leq \text{ulp}(2^{14}) = 2^{15-p}$. If $u$ straddles* 0 *then $|\text{mid}(u)| \leq \text{rad}(u)$, hence the true upper endpoint is at most $2\,\text{rad}(u) \leq 2^{16-p}$ and its square root is at most $2^{8-p/2}$ — exactly the coded constant. (In the definitive script the branch is replaced by* `abs_upper()`*, an automatic outer bound with no ad-hoc constant; the present lemma covers the archived run.)*

## The two scaled functions

The certificate's precision rests on three one-line facts. *(a)* $A(z) = e^{-z}I_1(z)/z$ is strictly decreasing on $(0, \infty)$: by the classical lower bound $I_{\nu+1}(z)/I_\nu(z) \geq z/(\nu + 1 + \sqrt{(\nu+1)^2 + z^2})$, valid for $z > 0$ and $\nu \geq 0$ (Amos [1]; both members of the classical two-sided pair are proved in [2], Theorem 1), taken at $\nu = 0$, together with $\sqrt{1 + z^2} \leq z + 1$ (square both sides), $I_1/I_0 > z/(z + 2)$, which is exactly $(\log A)' = I_0/I_1 - 1 - 2/z < 0$; hence $A$ is enclosed over any $z$-interval by its two endpoint values. *(b)* The kernel–score product collapses algebraically, $K\,\partial_t \log K = 32\beta^3 \cos(t/4)(\cos(t/4))'(1 - P - Q)\,H_B(z)$ (the factor $I_1(z)/z$ cancels between $K$ and the denominator of the regular score), so the covariance integrands involve only $B(z) = e^{-z}H_B(z)$. *(c)* $B$ is enclosed *unconditionally* by $[e^{-z_{\text{hi}}}H_B(z_{\text{lo}}),\, e^{-z_{\text{lo}}}H_B(z_{\text{hi}})]$, since $e^{-z}$ decreases and $H_B$ (a positive-coefficient entire series) increases; after centering, the $B$-terms are subdominant and this crude-but-safe bound suffices. Together with the exact linear-exponential cell integrals, every source of exponential interval growth is either integrated analytically or carried by a monotone one-dimensional profile — the certificate is structure, not brute force.

## 3 The spectral framework

*Convention* 22 (normalizations)*.* On $(0,\pi)$ define the kernel, entry law, and smoothing operator

$$q_1(u,v) = 2\, e^{\beta\cos u\cos v}\sinh(\beta\sin u\sin v) \;=\; 4\sum_{m\geq 1} I_m \sin(mu)\sin(mv), \tag{3}$$

$$s_1(u) = \tfrac{\beta}{2}\sin u\, e^{\beta\cos u} \;=\; \sum_{m\geq 1} m\, I_m \sin(mu), \tag{4}$$

$$(Qf)(v) = \frac{1}{2\pi}\int_0^\pi q_1(v,u) f(u)\, du, \tag{5}$$

so that $Q\sin(m\cdot) = I_m\sin(m\cdot)$. Both closed forms are classical consequences of the generating function $e^{\beta\cos\theta} = I_0 + 2\sum_{m\geq 1} I_m\cos(m\theta)$; (4) follows by differentiation. All ten verification artifacts in the repository use exactly these normalizations.

Since $q_1 > 0$ on $(0,\pi)^2$ and $s_1 > 0$ on $(0,\pi)$, the walk with step kernel $q_1$ killed on $\{0,\pi\}$ is well defined; $H(\psi) = \beta\sin(\psi/2) I_1(2\beta\cos(\psi/2))$ and $K_2(\varphi) = I_0(2\beta\cos(\varphi/2))$ appear below. Note $F_B = Q^3 s_1$ and, by the sine orthogonality computation of Lemma 29, $F_A$ is the companion cosine-weighted integral.

## 4 Theorem A: positivity of $F_B$

**Theorem 23** (exact)**.** *$F_B(t) > 0$ for all $t \in (0,\pi)$ and all $\beta > 0$.*

*First proof (kernel positivity).* $F_B = Q^3 s_1$ by (5) and $b_m = mI_m^4$. Each application of $Q$ integrates a positive function against the strictly positive kernel (3); $s_1 > 0$ on $(0,\pi)$. Hence $F_B > 0$ pointwise. □

*Second proof (symmetric-decreasing convolution).* With $K_2(\varphi) = I_0(2\beta\cos(\varphi/2))$ and $P_4 = K_2 * K_2$ (circle convolution; Section 7), $F_B = -\frac{1}{2}P_4'$. The function $K_2$ is even and *strictly decreasing in $|\varphi|$ on $[0,\pi]$* ($I_0$ strictly increasing, $\cos(\varphi/2)$ strictly decreasing). By the layer-cake representation, $K_2 = \int_0^\infty \mathbf{1}_{A_\lambda}\, d\lambda$ with $A_\lambda$ centered arcs whose half-lengths $a(\lambda)$ fill $(0,\pi)$ continuously (strict decrease). For centered arcs, $(\mathbf{1}_A * \mathbf{1}_B)(t) = |A\cap(t+B)|$ is even and nonincreasing in $|t|$ on $[0,\pi]$, and *strictly* decreasing on $\big||a-b|, \min(a+b, 2\pi-a-b)\big|$. Hence $P_4(t) = \iint |A_\lambda \cap (t+A_\mu)|\, d\lambda\, d\mu$ is even, nonincreasing on $[0,\pi]$, and strictly decreasing on $(0,\pi)$ because for every $t\in(0,\pi)$ a positive measure of pairs $(\lambda,\mu)$ has $|a(\lambda)-a(\mu)| < t < \min(a+b, 2\pi-a-b)$ (the half-lengths fill $(0,\pi)$). At the derivative level: each overlap $t\mapsto |A_\lambda\cap(t+A_\mu)|$ is Lipschitz with a.e. derivative $-1$ in the strict regime and 0 otherwise (on $[0,\pi]$); since $P_4$ is analytic, dominated convergence gives $P_4'(t) = \iint \partial_t|A_\lambda\cap(t+A_\mu)|\, d\lambda\, d\mu \leq -\mu\{(\lambda,\mu) : \text{strict at } t\} < 0$ pointwise on $(0,\pi)$. Thus $F_B = -\frac{1}{2}P_4' > 0$ on $(0,\pi)$. □

## 5 Theorem B: ratio monotonicity for $0 < \beta \leq 3$

**Lemma 24** (exact; Wronskian identity)**.** *For real arrays with $\sum m(|a_m| + |b_m|) < \infty$,*

$$W := 2(F_A' F_B - F_A F_B') = \sum_{1\leq m<n} c_{mn}\, T_{mn}(t), \qquad c_{mn} = a_m b_n - a_n b_m, \tag{6}$$

*where $T_{mn}(t) = (m-n)\sin((m+n)t) + (m+n)\sin((n-m)t)$.*

*Proof.* Product-to-sum: $T_{mn} = 2[m\cos(mt)\sin(nt) - n\sin(mt)\cos(nt)] =: 2G(m,n)$, antisymmetric with $G(n,n) = 0$; since $c_{mn}$ is antisymmetric, $\sum_{m<n} c_{mn}2G = 2\sum_{m,n} a_m b_n G(m,n) = 2[F_A' F_B - F_A F_B']$ (first recorded in [7]). ☐

The coefficient input is the weighted Turán-type lemma (Appendix A, self-contained): $\varphi_m = a_m/b_m$ is strictly increasing in $m \geq 1$ for every $\beta > 0$, hence *every minor is negative*: $c_{mn} < 0$ for $m < n$ (exact).

**Lemma 25** (exact; the pair identities and bounds)**.** *For $t \in (0,\pi)$, with $p = n-m$, $q = n+m$:*

*(a) $T_{12} = 4\sin^3 t$; $T_{13} = 16\sin^3 t\cos t$; $T_{24} = 8\sin^3(2t)$, so $|T_{13}| \leq 16\sin^3 t$ and $|T_{24}| \leq 64\sin^3 t$.*

*(b) $T_{m,m+1} \geq 0$ for every $m \geq 1$.*

*(c) $|T_{mn}| \leq \dfrac{\pi^3}{48}\, pq(q^2+p^2)\sin^3 t$ for all $m < n$.*

*Proof.* (a) $3\sin u - \sin 3u = 4\sin^3 u$ gives $T_{12}$ (at $u = t$) and $T_{24}$ (at $u = 2t$); $T_{13} = 4\sin 2t - 2\sin 4t = 4\sin 2t(1-\cos 2t) = 16\sin^3 t\cos t$. (b) $T_{m,m+1} = (2m+1)\sin t - \sin((2m+1)t) \geq 0$ since $|\sin(qt)| \leq q\sin t$ on $(0,\pi)$ (induction on $q$). (c) $T_{mn}$ and its first two derivatives vanish at $t = 0$ and $t = \pi$ (the parities of $m \pm n$ agree), and $|T_{mn}'''| \leq pq^3 + qp^3 = pq(q^2+p^2)$; hence $|T_{mn}(t)| \leq \frac{pq(q^2+p^2)}{6}\min(t,\pi-t)^3$, and $\sin t \geq \frac{2}{\pi}\min(t,\pi-t)$ gives the claim. ☐

**Theorem 26** (exact modulo the certified computation `certify_thmB.py`)**.** *For $0 < \beta \leq 3$: $W < 0$ on $(0,\pi)$, hence $E' < 0$ on $(0,\pi)$.*

*Proof.* Drop the adjacent pairs $m \geq 2$ (each contributes $c_{m,m+1}T_{m,m+1} \leq 0$ by Lemma 25(b) and $c < 0$); keep $T_{12}$, $T_{13}$, $T_{24}$ exactly and majorize every other pair by Lemma 25(c). Pointwise in $t$,

$$\frac{W(t)}{\sin^3 t} \;\leq\; \mathrm{Crit}(\beta) := -4|c_{12}| + 16|c_{13}| + 64|c_{24}| + \frac{\pi^3}{48}\sum_{\substack{p\geq 2\\ (m,n)\neq(1,3),(2,4)}} pq(q^2+p^2)\,|c_{mn}|.$$

The certificate `certify_thmB.py` (Section 10) encloses Crit in interval arithmetic over adaptive $\beta$-boxes covering $[1/20, 3]$ ($I_m$ increasing in $\beta$ gives rigorous boxes; cutoff $M = 60$ with the tail itself computed in intervals and closed by a derived, per-box-asserted geometric ratio) and certifies $\mathrm{Crit} < 0$ on all of $[1/20, 3]$; the script then re-certifies *every* box at precision +70 before reporting success (the box count is invocation- and implementation-dependent; the independent Arb implementation `certify_thmB_arb.py` certifies the same coverage with 86 boxes and a full stability pass, transcript archived). For $0 < \beta \leq 1/20$, Lemma 27 applies. ☐

**Lemma 27** (exact; small $\beta$)**.** *For $0 < \beta \leq 1/20$: $\mathrm{Crit}(\beta) < 0$.*

*Proof.* Write $\kappa = \beta/2 \leq 1/40$, $L_m = \kappa^m/m!$, $U_m = e^* L_m$ with $e^* = e^{\beta^2/4} \leq e^{1/1600} < 1.001$; then $L_m \leq I_m \leq U_m$ (positive series). *Leading minor:* $a_2 b_1 \geq (I_1^2 I_2^2)(I_1^4) \geq L_1^6 L_2^2 = \kappa^{10}/4$ and $a_1 b_2 = (2I_1^2 I_2^2)(2I_2^4) \leq 4U_1^2 U_2^6 = (e^*)^8\kappa^{14}/16$, so

$$|c_{12}| \;\geq\; \frac{\kappa^{10}}{4}\Big(1 - \frac{(e^*)^8\kappa^4}{4}\Big) \;\geq\; \frac{\kappa^{10}}{4}\,(1 - 10^{-6}).$$

*The three visible positive terms:* $a_3 \le I_3^2 \cdot (2I_2^2 + 4I_4^2) \le (e^*)^4\kappa^{10}/24 \cdot (1 + 10^{-4})$ and $b_1 \le (e^*)^4\kappa^4$ give $16|c_{13}| \le 16(a_3b_1 + a_1b_3) \le 0.70\,\kappa^{14}$; $64|c_{24}| \le 64(a_4b_2 + a_2b_4) \le 10^{-3}\kappa^{14}$ (both products have $\kappa$-degree $\ge 22$); the largest remaining weighted pair is $(1,4)$: $\frac{\pi^3}{48} \cdot 510 \cdot (a_4b_1 + a_1b_4) \le 0.2\,\kappa^{18} \le 10^{-6}\kappa^{14}$. *All other pairs* (those with $s := m + n \ge 6$, $p \ge 2$): using $a_nb_m + a_mb_n \le 2mn(e^*)^8\kappa^{4s-2}[((m-1)!m!)^{-2}n!^{-4} + ((n-1)!n!)^{-2}m!^{-4}] \le s^2(e^*)^8\kappa^{4s-2}$ (the factorial bracket is $\le 2$ and $2mn \le s^2/2$), $pq(q^2 + p^2) \le 2s^4$, and at most $s$ pairs per level,

$$\frac{\pi^3}{48}\sum_{s\ge6}(\text{level } s) \le \frac{\pi^3}{48}(e^*)^8\sum_{s\ge6}2s^7\kappa^{4s-2} \le 0.66\,\kappa^{22}\cdot 6^7\sum_{j\ge0}[(7/6)^7\kappa^4]^j \le 3.7\times10^5\,\kappa^{22} \le 10^{-7}\kappa^{14},$$

where the level ratio $s \mapsto s+1$ is at most $(7/6)^7\kappa^4 < 10^{-5} < \frac{1}{2}$ (so the geometric closure is valid), and $\kappa^8 \le 40^{-8}$ absorbs the constant. Collecting:

$$\text{Crit} \le -\kappa^{10}(1 - 10^{-6}) + 0.71\,\kappa^{14} = -\kappa^{10}(1 - 10^{-6} - 0.71\,\kappa^4) < 0,$$

since $\kappa^4 \le 40^{-4} < 4\cdot10^{-7}$. (All numbered constants above are crude upper counts; each step loses factors, none gains.) $\square$

*Remark* 28 (verified; the sharp family and saturation). The sharper per-pair bound $|T_{mn}| \le \frac{pq(q^2-p^2)}{6}\sin^3 t$ and the adjacent floor $T_{m,m+1} \ge 2m\sin^3 t$ (note the sign: the floor is a *lower* bound on a positive quantity) are verified far beyond the range used ($q \le 151$; 13 pairs to $(10,30)$, max ratio 1.0) and would extend Theorem 26 to $\beta \approx 3.5$; their per-pair certification is a scheduled upgrade. Beyond that the decomposition is *saturated*: with the exact far block (cancellation included) the criterion fails at $\beta = 6$ by $+2.2\times10^{14}$ while the true $W/\sin^3 t$ remains negative ($-6.9\times10^{12}$) — a verified adversarial finding about the method, and the reason the large-$\beta$ closure uses different machinery (Sections 7–9).

# 6 The bridge structure of $E$ (exact)

Let $k = q_1 \circ q_1$ denote the two-step kernel, $k(\psi, t) = 4\sum_{m\ge1} I_m^2 \sin(m\psi)\sin(mt)$, and define

$$\omega_t(d\psi) \;\propto\; (Qs_1)(\psi)\,k(\psi,t)\,d\psi \qquad \text{on } (0,\pi), \tag{7}$$

the law of the time-2 position (midpoint) of the four-step bridge entering from 0 with law $s_1$ and conditioned to end at $t$.

**Lemma 29** (exact)**.** *$\int_0^\pi (Qs_1)\,k(\cdot,t)\,d\psi \;=\; 2\pi F_B(t)$ and $\int_0^\pi \cos\psi\,(Qs_1)\,k(\cdot,t)\,d\psi \;=\; \pi F_A(t)$. Hence, as an identity,*

$$E(t) \;=\; \mathbb{E}_{\omega_t}[\cos\psi] \;=\; \frac{F_A(t)}{2F_B(t)}.$$

*Proof.* $(Qs_1)(\psi) = \sum mI_m^2\sin(m\psi)$. By sine orthogonality the first integral picks the diagonal, giving $2\pi\sum mI_m^4\sin(mt)$. In the second, $\cos\psi\,\sin(m\psi)\sin(n\psi)$ integrates to $\frac{\pi}{4}(\delta_{n,m+1} + \delta_{n,m-1})$, which produces exactly the weights $(n-1)I_{n-1}^2 + (n+1)I_{n+1}^2$ of $a_n$. $\square$

**Corollary 30** (exact; $|E| < 1$)**.** *For all $t \in (0,\pi)$ and $\beta > 0$, $|E(t,\beta)| < 1$. Indeed $q_1(u,v) = 2e^{\beta\cos u\cos v}\sinh(\beta\sin u\sin v) > 0$ on $(0,\pi)^2$ and $s_1 > 0$, so $(Qs_1) > 0$ and $k = q_1 \circ q_1 > 0$ on $(0,\pi)$; hence $\omega_t$ in (7) is a genuine probability measure and $E = \mathbb{E}_{\omega_t}[\cos\psi] \in (-1,1)$.*

**Lemma 31** (exact; entry law)**.** *$\partial_t k(0^+,\psi) = 2H(\psi)$, where $H(\psi) = \beta\sin(\psi/2)\, I_1(2\beta\cos(\psi/2))$.*

*Proof.* Differentiate in $\psi$ the squared Graf identity $I_0^2 + 2\sum_{m\ge1} I_m^2 \cos(m\psi) = I_0(2\beta\cos(\psi/2))$ [4, §10.23(ii)] and compare with $\partial_t k(0^+,\psi) = 4\sum m I_m^2 \sin(m\psi)$. □

**Proposition 32** (certified negative; no monotone full-path coupling)**.** *At $\beta = 20$ the time-3 marginal $\rho_t(u) \propto (Q^2 s_1)(u)\, q_1(u,t)$ is not stochastically monotone in $t$: with $a = \frac{4802}{10^4}$, $t_1 = \frac{29621}{10^4}$, $t_2 = \frac{30070}{10^4}$, the tail masses satisfy $T(t_2,a) < T(t_1,a)$, with certified enclosure*

$$T(t_2,a) - T(t_1,a) \in [-2.5299150980081786690028107,\\ -2.5299150980081786690028074] \times 10^{-10}.$$

*Consequently no monotone coupling of full bridge paths exists at $\beta = 20$. The certificate uses exact termwise tails ($J(m,n,a) = \int_a^\pi \sin mu \sin nu\, du$ in closed form: no quadrature error), positive-series Bessel enclosures with geometric tails, explicit truncation bounds, nested passes* $(M, \mathrm{prec}) = (120, 350) \subset (140, 500)$*, and two independent interval implementations (`mpmath.iv`; Arb via `python-flint`).*

**Proposition 33** (certified negatives; the weight–kernel matrix)**.** *At $\beta = 20$, for every entry weight $Q^{k-1}s_1$ ($k = 1,2,3,4$) the marginal one $q_1$-step from the moving endpoint fails stochastic monotonicity, with certified witnesses sharing $(t_1,t_2) = (3, 61/20)$: $k = 1$ (two-step bridge midpoint, the minimal counterexample), $a = 9/10$, margin $\approx -1.34\times10^{-2}$; $k = 2$, $a = 3/4$, $\approx -1.80\times10^{-6}$; $k = 4$, $a = 3/10$, $\approx -4.75\times10^{-13}$ ($k = 3$ is Proposition 32). Two smoothing steps repair all tested weights (verified); one never does. The statement is relative to the natural weight family: $k_2$ possesses its own total-positivity corner ($\beta \gtrsim 8$) reachable by two-atom adversarial weights.*

**Lemma 34** (exact; the 3/2 mechanism)**.** $\lim_{u\downarrow0} \dfrac{\partial_u\partial_t \log q_1(u,t)}{\beta \sin t\; u} = 1 + \dfrac{2\beta}{3}\cos t$*. For $t > \pi/2$ the corner sign change of $\partial_u\partial_t \log q_1$ occurs exactly when $\beta|\cos t| > 3/2$ — the same threshold as the $TP_2$ failure of $q_1$.*

*Proof.* $\partial_t \log q_1 = -\beta\sin t\cos u + \beta\cos t \sin u \coth(\beta \sin t \sin u)$ (direct differentiation of (3)); expand $\coth z = z^{-1} + z/3 + O(z^3)$ and differentiate in $u$; the stated limit is a one-line computation (symbolic residual zero). □

*Remark* 35 (verified; the mechanism reading). The identification of this local defect as *the* engine of Propositions 32–33 — sine-type entry weights, vanishing linearly at 0 with mass on both sides of the non-monotone region, flip the covariance $\mathrm{Cov}_{\rho_t}(\mathbf{1}_{u\ge a}, \partial_t \log q_1)$, while the flat weight does not — is supported by the full weight–kernel matrix and the $\beta$-scan of the violations (onset at $\beta \gtrsim 12$, consistent with the threshold), but is a verified reading, not a theorem.

# 7 The single-Bessel reduction and the master formula

**Lemma 36** (exact; Neumann product formulas)**.** *For $x, y \ge 0$, with $w = \sqrt{x^2+y^2+2xy\cos\alpha}$,*

$$I_0(x)I_0(y) = \frac{1}{2\pi}\int_0^{2\pi} I_0(w)\, d\alpha, \qquad I_0(x)I_1(y) = \frac{1}{2\pi}\int_0^{2\pi} I_1(w)\,\frac{y + x\cos\alpha}{w}\, d\alpha.$$

*Proof.* The first is the integrated Graf/Neumann addition theorem [4, §10.23(ii)]; the second is its $\partial_y$-derivative. □

*Remark* 37 (signs). Both formulas extend to all real $x, y$, as used below (on $[-\pi,\pi]^2$ one has $c_1 = \cos(\varphi/2) \ge 0$ but $c_2$ may be negative): $w(x,-y,\alpha) = w(x,y,\pi-\alpha)$, and the substitution $\alpha \mapsto \pi - \alpha$ over a full period maps the first identity to itself (both sides even in $y$) and the second to its negative on both sides simultaneously ($I_1$ odd; $(-y + x\cos(\pi-\alpha)) = -(y + x\cos\alpha)$).

With $K_2(\varphi) = I_0(2\beta\cos(\varphi/2)) = \sum_{m\in\mathbb{Z}} I_m^2 e^{im\varphi}$ and $P_4 = K_2 * K_2$ (circle convolution), $P_4 = \sum_m I_m^4 e^{imt}$ and $F_B = -\frac{1}{2}P_4'$. Applying Lemma 36 to both convolution factors:

**Lemma 38** (exact; master representation)**.** *Let* $s = \varphi - t/2$, $c_1 = \cos(\varphi/2)$, $c_2 = \cos((t-\varphi)/2)$, $P = \sin^2(s/2)$, $Q = \sin^2(\alpha/2)$, *and*

$$R^2 = c_1^2 + c_2^2 + 2c_1c_2\cos\alpha = 4\big[\cos^2(t/4)\,(1-P-Q) + PQ\big]. \tag{8}$$

*Then*

$$F_B(t) = \frac{\beta\sin(t/2)}{8\pi^2}\iint_{[-\pi,\pi]^2} \frac{I_1(2\beta R)}{R}\big[\cos^2(s/2) - \sin^2(\alpha/2)\big]\,ds\,d\alpha, \tag{9}$$

*and, under the common positive kernel* $K = I_1(2\beta R)/R$,

$$E(t) = \frac{\big\langle \cos(t/2)\cos 2s + \cos\alpha\,[\cos(t/2)\cos s - \sin^2 s]\big\rangle}{\big\langle \cos s + \cos\alpha \big\rangle}, \tag{10}$$

*where* $\langle\cdot\rangle$ *denotes the* $K\,ds\,d\alpha$*-integral over* $[-\pi,\pi]^2$*. The prefactor collapse behind* (9) *is the sum-to-product symmetrization* $\frac{1}{2}[\sin(\frac{t-\varphi}{2})(c_2 + c_1\cos\alpha) + (\varphi \leftrightarrow t-\varphi)] = \frac{1}{2}\sin(t/2)(\cos s + \cos\alpha)$, *and* $\cos s + \cos\alpha = 2[\cos^2(s/2) - \sin^2(\alpha/2)]$.

*Proof.* (i) The squared Graf identity gives $K_2(\varphi) = \sum_m I_m^2 e^{im\varphi}$; convolution multiplies Fourier coefficients, so $P_4 = \sum_m I_m^4 e^{im\varphi}$ and, termwise (super-exponential decay), $F_B = -\frac{1}{2}P_4'$. (ii) Insert Lemma 36 (first formula) with $x = 2\beta c_1$, $y = 2\beta c_2$ into $P_4(t) = \frac{1}{2\pi}\int K_2(\varphi)K_2(t-\varphi)\,d\varphi$: a two-dimensional integral of $I_0(2\beta R)$ with $R$ as in (8); the $(P,Q)$ form of $R^2$ is product-to-sum. (iii) Differentiate under the integral (smooth integrand, compact domain): $\partial_t I_0(2\beta R) = I_1(2\beta R)\cdot 2\beta\,\partial_t R$ and $2R\,\partial_t R = 2(c_2 + c_1\cos\alpha)\,\partial_t c_2$ with $\partial_t c_2 = -\frac{1}{2}\sin((t-\varphi)/2)$. (iv) Symmetrize over $\varphi \leftrightarrow t-\varphi$ ($R$ invariant): with $\sin(x/2)\cos(x/2) = \frac{1}{2}\sin x$ and the angle-addition formula,

$$\begin{aligned}\tfrac{1}{2}\big[\sin(\tfrac{t-\varphi}{2})(c_2 + c_1\cos\alpha) + \sin(\tfrac{\varphi}{2})(c_1 + c_2\cos\alpha)\big] &= \tfrac{1}{4}[\sin(t-\varphi) + \sin\varphi] + \tfrac{1}{2}\sin(\tfrac{t}{2})\cos\alpha \\ &= \tfrac{1}{2}\sin(\tfrac{t}{2})(\cos s + \cos\alpha),\end{aligned}$$

using $\sin(t-\varphi) + \sin\varphi = 2\sin(t/2)\cos s$ and $\sin(\frac{t-\varphi}{2})c_1 + \sin(\frac{\varphi}{2})c_2 = \sin(t/2)$. Collecting constants yields (9). The same substitution in the companion representation of $F_A$ (Lemma 29 with Lemma 36, second formula) and cancellation of the common factor $\beta\sin(t/2)$ yields (10). As an end-to-end check, (9) and (10) match the quartic series with ratio 1.000000000000 at $t \in \{0.4, 1.1, 2.3, 2.9\}$, $\beta = 4$. $\square$

**Lemma 39** (exact; geometry of the $(P,Q)$ square, $0 < t < \pi$)**.** *On* $[0,1]^2$*: the main saddle is* $(0,0)$ *with* $R = 2\cos(t/4)$*; the secondary saddle is* $(1,1)$ *with* $R = 2\sin(t/4)$ *and gap* $2(\cos(t/4) - \sin(t/4)) > 0$ *degenerating only at* $t = \pi$*; the zero set* $\{R = 0\}$ *consists of exactly the two corners* $(1,0)$ *and* $(0,1)$*, and at both of them* both *brackets of* (9) *and* (10) *vanish, so the singular points of* $K$ *are switched off (the extension* $I_1(z)/z \to \frac{1}{2}$ *makes* $K$ *continuous with value* $\beta$*). At* $t = 0$ *the zero set degenerates to the edges* $P = 1$*,* $Q = 1$*. The saddle deficit is the exact sum of two squares*

$$4\cos^2(t/4) - R^2 = 4\cos^2(t/4)\sin^2(s/2) + 4c_1c_2\sin^2(\alpha/2), \qquad c_1c_2 = \cos^2(t/4) - \sin^2(s/2), \tag{11}$$

*with sign region* $c_1c_2 \ge 0 \iff |s| \le \pi - t/2$*.*

*Proof.* (8) and (11) are product-to-sum computations (symbolic zeros). $R^2 \geq (|c_1| - |c_2|)^2$ with equality forcing $|c_1| = |c_2|$ (i.e. $s = 0$ or $s = \pm\pi$) and $\cos\alpha = \mp\operatorname{sign}(c_1 c_2)$, which lands exactly on $(P, Q) \in \{(0, 1), (1, 0)\}$. Bracket vanishing at both points is direct substitution. The Jacobian of $(s, \alpha) \mapsto (P, Q)$ carries the arcsine weight $ds\, d\alpha = dP\, dQ/\sqrt{P(1-P)Q(1-Q)}$ (per quadrant), singular at all four corners; near the main saddle one therefore works in $(s, \alpha)$, and in $(P, Q)$ only for the global geometry. □

**Lemma 40** (exact; saddle cancellations). *Write $N, D$ for numerator and denominator brackets of (10), $C = \cos(t/2)$, $c_0 = \cos(t/4)$. At the saddle $(s, \alpha) = (0, 0)$:*

$$\frac{N}{D}\Big|_{(0,0)} = C, \qquad N_{\alpha\alpha} - C\, D_{\alpha\alpha} = 0, \qquad N_{ss} - C\, D_{ss} = -4C - 2,$$

*and for $N_t = -\frac{1}{2}\sin(t/2)[\cos 2s + \cos\alpha\cos s]$:*

$$N_{t,\alpha\alpha} - m\, D_{\alpha\alpha} = 0 \quad (m = -\tfrac{1}{2}\sin(t/2)), \qquad N_{t,ss} - m\, D_{ss} = 2\sin(t/2).$$

*Also exact: $2C + 1 = 4c_0^2 - 1$ and $C\, c(t) = 2c_0 - 1/(2c_0)$ for $c(t) = (4c_0^2 - 1)/(2c_0 C)$.*

**Proposition 41** (verified; the second-order laws). *The frozen-kernel expansion built on Lemma 40 (with $\langle s^2 \rangle = 1/(\beta c_0) + O(\beta^{-2})$ from (11)) predicts*

$$E = C\Big(1 - \frac{c(t)}{\beta}\Big) + O(\beta^{-2}), \qquad E' = -\tfrac{1}{2}\sin(t/2) + \frac{\sin(t/4)(4c_0^2 + 1)}{8\beta c_0^2} + O(\beta^{-2}). \tag{12}$$

*The coefficient identities are exact (Lemma 40); the* laws with remainder *are verified, not yet certified: measured remainders are $\approx 0.15/\beta^2$ with true convergence at fixed $t$ (e.g. $\beta^2$-scaled residues 0.1776/0.1767/0.1762 at $t = 1$, $\beta = 30/60/120$), and $\beta\,(E' + \frac{1}{2}\sin(t/2)) \to \sin(t/4)(4c_0^2 + 1)/(8c_0^2)$ at $t = 1, 2$. The remainder control is exactly what the closure map owes. Note the $1/\beta$ correction to $E'$ is* positive*: it opposes the target inequality and must be beaten, not used. The $\alpha$-cancellations imply every kernel correction multiplies a vanishing fluctuation and enters at $O(\beta^{-2})$ only.*

*Remark* 42 (exact decomposition of $E'$). With the normalized positive measure $d\mu \propto K\, ds\, d\alpha$, $E' = \langle N_t \rangle / \langle D \rangle + \langle (N - ED)\, \partial_t \log K \rangle / \langle D \rangle$, $D_t = 0$, and the regular form $\partial_t \log K = 4cc'(1 - P - Q)\,[zI_0(z)/I_1(z) - 2]/R^2$, $z = 2\beta R$, extends continuously by $4\beta^2 cc'(1 - P - Q)$ at $R = 0$. Verified split at $(\beta, t) = (5, 1.3)$: $-0.238212 - 0.024552 = -0.262764$ = the finite difference exactly. Bounds for $E'$ are taken from this exact quotient — never by differentiating an estimate of $E$.

## 8 The $\pi$-endpoint package

**Lemma 43** (exact; telescoping). *With $a_m$ as in (1):*

$$A_\infty := \sum_{m \geq 1} (-1)^{m+1} m\, a_m = 0, \qquad \sum_{m \geq 1} (-1)^{m+1} m^3 a_m = -\sum_{m \geq 1} (-1)^{m+1} m(m+1)(2m+1)\, I_m^2 I_{m+1}^2.$$

*Hence $F_A(\pi - \varepsilon) = c_3\, \varepsilon^3 + O(\varepsilon^5)$ with*

$$c_3 = \frac{1}{6} \sum_{m \geq 1} (-1)^{m+1} m(m+1)(2m+1)\, I_m^2 I_{m+1}^2. \tag{13}$$

*Proof.* Split $m^r a_m = m^r(m-1)I_{m-1}^2 I_m^2 + m^r(m+1)I_m^2 I_{m+1}^2$ and shift $m \to m+1$ in the first sum: for $r=1$ the two weights cancel ($m(m+1)-m(m+1)=0$); for $r=3$ they combine to $m(m+1)[m^2-(m+1)^2] = -m(m+1)(2m+1)$. Absolute convergence justifies the reindexing; $\sin(m(\pi-\varepsilon)) = (-1)^{m+1}\sin(m\varepsilon)$ expands the sine. $\square$

**Lemma 44** (exact; integral form and parity)**.** *Let* $g(u) = \frac{1}{2}I_1(2\beta\cos u) = \sum_{m\geq 0} I_m I_{m+1}\cos((2m+1)u)$ (addition theorem)*. Then*

$$c_3 = \frac{S_3 - S_1}{24}, \qquad S_1 = \frac{2}{\pi}\int_0^\pi g(\tfrac{\pi}{2}-u)\,g'(u)\,du, \quad S_3 = \frac{2}{\pi}\int_0^\pi g'(\tfrac{\pi}{2}-u)\,(-g''(u))\,du.$$

*Moreover* $g(-u) = g(u)$ *and* $g(\pi-u) = -g(u)$ *(parity of* $I_1$*), hence* $g'(\pi-u) = g'(u)$*,* $g''(\pi-u) = -g''(u)$*, and* both *integrands are invariant under* $u \mapsto \pi - u$*: the two halves* $[0,\pi/2]$ *and* $[\pi/2,\pi]$ *contribute exactly equally, and one may integrate a half range and double. In particular no cancellation between the two phase maxima* $u=\pi/4$ *and* $u=3\pi/4$ *is possible — exactly, not asymptotically.*

*Proof.* The expansion of $g$ is the Graf addition theorem with unit shift. Quarter turn: $\cos((2m+1)(\frac{\pi}{2}-u)) = (-1)^m\sin((2m+1)u)$, so $g(\frac{\pi}{2}-u) = \sum_m (-1)^m I_m I_{m+1}\sin((2m+1)u)$ while $g'(u) = -\sum_k (2k+1)I_k I_{k+1}\sin((2k+1)u)$ and $g''(u) = -\sum_k (2k+1)^2 I_k I_{k+1}\cos((2k+1)u)$, whence $g'(\frac{\pi}{2}-u) = -\sum_k (-1)^k(2k+1)I_k I_{k+1}\cos((2k+1)u)$. Orthogonality of $\{\sin((2m+1)u)\}$ and $\{\cos((2m+1)u)\}$ on $[0,\pi]$ (norm $\pi/2$) gives $S_1 = -\sum_m (-1)^m(2m+1)(I_m I_{m+1})^2$ and $S_3 = -\sum_m(-1)^m(2m+1)^3(I_m I_{m+1})^2$, so

$$\begin{aligned}\frac{S_3-S_1}{24} &= \sum_{m\geq 0}(-1)^{m+1}\frac{(2m+1)^3-(2m+1)}{24}(I_m I_{m+1})^2 \\ &= \frac{1}{6}\sum_{m\geq 1}(-1)^{m+1}m(m+1)(2m+1)I_m^2 I_{m+1}^2 = c_3,\end{aligned}$$

using $(2m+1)^3-(2m+1) = 4m(m+1)(2m+1)$ and the vanishing of the $m=0$ term. The parity claims follow from $I_1(-z) = -I_1(z)$ applied to $g(\pi-u) = \frac{1}{2}I_1(-2\beta\cos u)$, and by differentiating the two symmetry relations. $\square$

**Proposition 45** (verified; prefactor law)**.** $c_3 > 0$ *for* $\beta \in \{1,5,20,40,80,120\}$ *at cancellation-adjusted precision* (the alternating sum cancels at scale $e^{-(4-2\sqrt{2})\beta}$; evaluations use dps $\geq 0.6\beta + 50$)*, and*

$$c_3 \;\sim\; A\,\beta^{3/2}e^{2\sqrt{2}\beta}, \qquad A = \frac{1}{24\cdot 2^{1/4}\pi^{3/2}} = 0.0062922\ldots,$$

*with measured ratios* 0.0058646, 0.0060049, 0.0060996, 0.0061475 *at* $\beta = 40, 60, 90, 120$ *and* $O(1/\beta)$ *gaps. The certified large-*$\beta$ *branch (pending) must produce the signed prefactor with an explicit* $B$*:* $I_{\text{half}} \geq \frac{A}{2}\beta^{3/2}e^{2\sqrt{2}\beta}(1-B/\beta) - T(\beta)$*,* $c_3 = 2I_{\text{half}}$*, read* $\beta^*$*, and close* $[15, B_{\text{fin}}]$*,* $B_{\text{fin}} = \max\{\beta^*, 10C\}$*, by intervals.*

## 9 The closure map and the candidate threshold

**Proposition 46** (verified truth map; all statements on the stated finite grids)**.** *Let* $g(t,\beta) = E' + \frac{1}{4}\sin(t/2)$*. At* $\beta = 15$*,* $g < 0$ *at every point of a* 13*-point grid spanning* $[0.05, \pi - 0.09]$ *(independently reproduced on a much finer mesh at* 110 *digits), with thinnest margin* $-0.00573$

*at* $t = 0.05$ *(on the seal window* $[0.05, \pi - 0.10]$*) and* $-0.00216$ *at* $\pi - 0.09$*; the crossing sits at* $\pi - t^* = 0.0890502225$*. At* $\beta = 12$ *the truth itself fails at* $\pi - 0.10$ *(*$g = +0.025$*); crossings scale as the mirror window:* $\pi - t^* = 0.1155, 0.0891, 0.0644$ *at* $\beta = 12, 15, 20$*, matching* $\sqrt{2}/\beta$*. A sweep* $\beta = 15 \dots 60$ *keeps the maximum at* $t = 0.05$*, drifting to* $-0.0061$*: strong uniformity evidence, not proof.*

**Conjecture 47** (the pending rows of the relay, quantified)**.** *Part* (ii) *of Conjecture 1 holds with the two-scale decomposition:* (a) $0 < \beta \leq 3$ *by Theorem 26;* (b) $\beta \in [3, 15]$ *by certified evaluation* (the holonomic structure of Section 10 makes the compact cheap)*;* (c) $\beta \geq 15$ *by three uniform blocks* (windows per the LIVE map of Section 2, superseding earlier drafts)*: left edge* $(0, 0.6]$ (*by the ratified sinc route of Remark 4: the exact sinc positivity of Lemma 5, the certified minor criterion with control of the high-frequency tail* $m+n > \pi/t$*, and the uniform splice overlap with the integrator strip extended down to* $t_*(\beta) \approx 0.45$)*; bulk* $[0.6, \pi - C_{\text{win}}/\beta]$ (*enclosure of* $\langle s^2 \rangle$ *and of the covariance of Remark 42, with the five prototype-to-certificate conversions*)*; right edge with* moving *boundary: assembly to* $\pi - C_{\text{win}}/\beta$*, then the quadratic regime of* $c_3$ *on* $[\pi - C_{\text{win}}/\beta, \pi]$ *with* $C_{\text{mirror}} \leq C_{\text{win}} \leq C_{\text{quad}}$ (*a fixed quadratic window is provably impossible:* $\kappa \approx 0.11\beta$ *grows while* $E(\pi - 0.10)$ *stays bounded; retention* $E/\kappa d^2$ *at* $d = 2/\beta$ *is* $0.896, 0.892, 0.890$ *at* $\beta = 20, 40, 80$ *— the validity radius scales as* $1/\beta$)*. The named constants still owed:* $B$*,* $\beta^*$*,* $C_{\text{quad}}$*,* $C_{\text{mirror}}$*, and the remainder function* $T(\beta)$*.*

The first-order competition is not the obstacle: the two-term law (12) requires only $\beta > (1 + 1/(4c_0^2))/c_0 \leq 3\sqrt{2}/2 \approx 2.121$; everything above that is rigor slack. The candidate $\beta_0 = 15$ is near truth-optimal for its edge window (Proposition 46), not padded.

## 10 Verification, certificates, and reproducibility

All numerical artifacts live in the repository

https://github.com/lluiseriksson/THE-ERIKSSON-PROGRAMME

(directories `scripts/`, `docs/`); the immutable commit hash for the version audited with this manuscript is recorded in the repository's hash registry (`docs/SURFACE-CLOSURE-NOTES.md`) and in the arXiv comment field on submission. Each certificate embeds its pinned statement and self-verifies on every execution.

*Certificates (interval arithmetic, outward rounding, nested or stability passes).* `certify_time3_negative.py` and its Arb twin (Proposition 32); `certify_bridge_matrix.py` and twin (Proposition 33); `certify_thmB.py` (Theorem 26: adaptive $\beta$-boxes covering $[1/20, 3]$; tails computed in intervals and closed at the level of *complete shells*: each pair maps into the next shell with term ratio $r$ derived from the termwise-exact $I_{n+1}(x) \leq x I_n(x)/(2(n+1))$, and the one new pair per shell is bounded by a double index shift, giving shell ratio $2r$, asserted $< \frac{1}{2}$ per box before the geometric closure; a built-in full stability pass at precision $+70$; the Arb twin `certify_thmB_arb.py` independently certifies the full coverage — 86 boxes, both passes, transcript archived; the `mpmath.iv` canonical transcript is committed on completion of its run, as recorded in the repository hash registry). Bessel enclosures use the positive series with an exact-integer stopping rule and geometric tails; floats appear only in loop-termination heuristics. Nested enclosures prevent silent numerical or truncation degradation; statement auditing and the independent Arb implementation guard against common-mode errors.

*Verified computations.* The master-formula and reduction identities (ratios 1.0 to 12 digits); the truth map of Proposition 46 (dps 80, confirmed independently at 110 digits); the $c_3$ ladder (Proposition 45); the term-split coefficients of (12) at $\beta = 60, 120, 240$.

*Precision policies* (each traceable to a caught artifact): dps $\geq 2.2\beta + 40$ for any evaluation with $t \geq 2$, $\beta > 30$ (mirror cancellation); dps $\geq 0.6\beta + 50$ for $c_3$-type alternating sums; series cutoffs scale with the maximal argument, and scaled-asymptotic evaluation of $e^{-z} I_n(z)$ is the default for $z > 35$; every measured coefficient carries a $\beta$-scaling sanity check; whatever can be computed from a positive integral representation is.

*Audit trail.* Fifteen numerical or formulation artifacts ("ghosts") were caught during fabrication and writing by the three-way review protocol; Appendix B tabulates them. We consider the table part of the result: it is the reproducibility record that makes the certificates credible.

*Machine-checked lemmas.* The unit-step and Turán-type coefficient inequalities behind Theorem 26 are formalized in Lean 4/Mathlib (`PhiMonotone.lean`, `FHBesselAmos.lean`, `WronskianIdentity.lean`; axiom oracle `[propext, Classical.choice, Quot.sound]`, no `sorry`). Mathlib has no Bessel functions: what is machine-checked is the ordered-field implication, with the analytic inputs (three-term recurrence [4, 10.29.1]; the Amos-type bound of [1, 2, 3], cited verbatim, shift $\nu \to \nu+1$ as in Appendix A) carried as named hypotheses. Correct claim everywhere: *machine-checked modulo classical Bessel inputs carried as named hypotheses.*

## Acknowledgements and provenance

The conjecture, the bridge reading, the reduction, and all certificates were produced inside an audit-first programme in which three independent reviewing voices verify every claim before acceptance; scores, objections, and the complete ghost ledger are archived with the repository history. Nothing in this paper rests on trust in any single computation: every load-bearing negative has two witnesses, and every identity marked exact has a symbolic-zero transcript.

## A The weighted Turán-type lemma, self-contained

**Theorem 48** (exact)**.** *For every $x > 0$ and every real $m \geq 1$, with $\varphi_m = a_m/b_m$ built from (1) at argument $x$: $\varphi_m < \varphi_{m+1}$. Equivalently, with $\rho_\mu = I_{\mu+1}/I_\mu$,*

$$\frac{1}{m}\Big(\frac{m-1}{\rho_{m-1}^2} + (m+1)\rho_m^2\Big) < \frac{1}{m+1}\Big(\frac{m}{\rho_m^2} + (m+2)\rho_{m+1}^2\Big).$$

*Proof.* Two classical inputs, valid for all real orders: the three-term recurrence $\rho_{m-1}^{-1} = \rho_m + 2m/x$, $\rho_{m+1} = \rho_m^{-1} - 2(m+1)/x$ [4, 10.29.1], and the Amos-type bound

$$\rho_m(x) < U_m(x) := \frac{x}{a + \sqrt{a^2 + x^2}}, \qquad a = m + \tfrac{1}{2},$$

proved best of its form for all real orders $\geq 0$ by Segura [2]; we use [3, Thm. 2], which bounds $I_\nu(x)/I_{\nu-1}(x)$ for $\nu \geq \frac{1}{2}$ by $x/(\nu - \frac{1}{2} + \sqrt{(\nu+\frac{1}{2})^2 + x^2}\,)$; setting $\nu = m+1$ bounds $\rho_m = I_{m+1}/I_m$ by $x/(m + \frac{1}{2} + \sqrt{(m+\frac{3}{2})^2 + x^2}\,) \leq x/(a + \sqrt{a^2 + x^2}\,)$ with $a = m + \frac{1}{2}$, which is the displayed bound (the last step uses $\sqrt{(m+\frac{3}{2})^2 + x^2} \geq \sqrt{a^2 + x^2}$). Substitute the recurrences, write

$u = \rho_m \in (0,1)$, $c = 1/x$, $S = u + 1/u$, $P = 1/u - u$; clearing the positive denominators, the claim becomes $\Delta_m > 0$ where the *exact factorization*

$$\frac{\Delta_m}{2m(m+1)} = (S - 3c)\big(P - (2m+1)c\big) + (2m+1)c^2$$

holds as an algebraic identity (expand and collect; the coefficients of $u^{\pm 2}$ are $\pm 2m(m+1)$, the linear-in-$c$ terms are $-4m(m+1)c[(m-1)u + (m+2)/u]$, and $(m-1)u + (m+2)/u = \frac{1}{2}[(2m+1)S + 3P]$, $1/u^2 - u^2 = PS$). The second factor is positive because $U_m$ is the positive root of $1/w - w = (2m+1)/x$ (the calibration $1/U_m - U_m = 2a/x$) and $w \mapsto 1/w - w$ is strictly decreasing, so $\rho_m < U_m$ gives $P > (2m+1)c$. The first factor is positive because $\sqrt{a^2 + x^2} > a$ gives $U_m < x/(2a)$, so $u < x/(2m+1) \leq x/3$ and $S > 1/u > (2m+1)/x \geq 3c$. The third term is positive. Strictness is strict throughout. (This is the proof of [6], reproduced so that Theorem 26 is self-contained; the ordered-field implication is machine-checked in `PhiMonotone.lean`, modulo the two classical inputs carried as named hypotheses.) □

# B Audit table (the ghost ledger)

Each row: what was wrong, how it was caught, and the rule it produced. All entries are documented with transcripts in `docs/SURFACE-CLOSURE-NOTES.md`.

| # | artifact | catch / rule produced |
|---|---|---|
| 1 | random $TP_2$ scan missed adversarial corners | targeted corner search; adversarial grids mandatory |
| 2 | verification cell with unbound parameter | explicit binding rule for inline prints |
| 3 | near-$\pi$ scan at low precision, false violation | precision rule dps $\geq 2.2\beta + 40$ for $t \geq 2$, $\beta > 30$ |
| 4 | stale-PDF claim misattributed between voices | provenance ledger: attribute to exact source |
| 5 | float64 CDF increments $\pm 11$ (impossible) | positive-representation preference |
| 6 | $t$=2, $\beta$=240 residue 0.363 vs 0.317 | boundary $t \geq 2$ included in the dps rule |
| 7 | $I_1$ series truncated at 60 terms, $z \leq 480$ | cutoffs scale with argument; scaled asymptotics for $z > 35$; $\beta$-scaling checks |
| 8 | $c_3$ sign flip at $\beta = 120$ (cancellation) | alternating-sum rule dps $\geq 0.6\beta + 50$ |
| 9 | $R = 0$ corner misplaced at $(1, 1)$ | complete critical-point census required in prose too |
| 10 | one-saddle prefactor off by exactly 2 | census rule; the factor 2 later proved an exact parity |
| 11 | $I_1$ lower bound dropped the negative tail | sign audits are a separate mandatory pass |
| 12 | fixed quadratic window at $\pi$ was provably false | interrogate obligations before attempting them |
| 13 | Thm. A second proof: rearrangement misquoted | layer-cake arc proof substituted (this revision) |
| 14 | Lemma sign error: floor stated as upper bound | corrected; per-pair statements now carry signs explicitly |
| 15 | "exact" labels on verified inequalities | tricotomy re-audit of every label (this revision) |

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